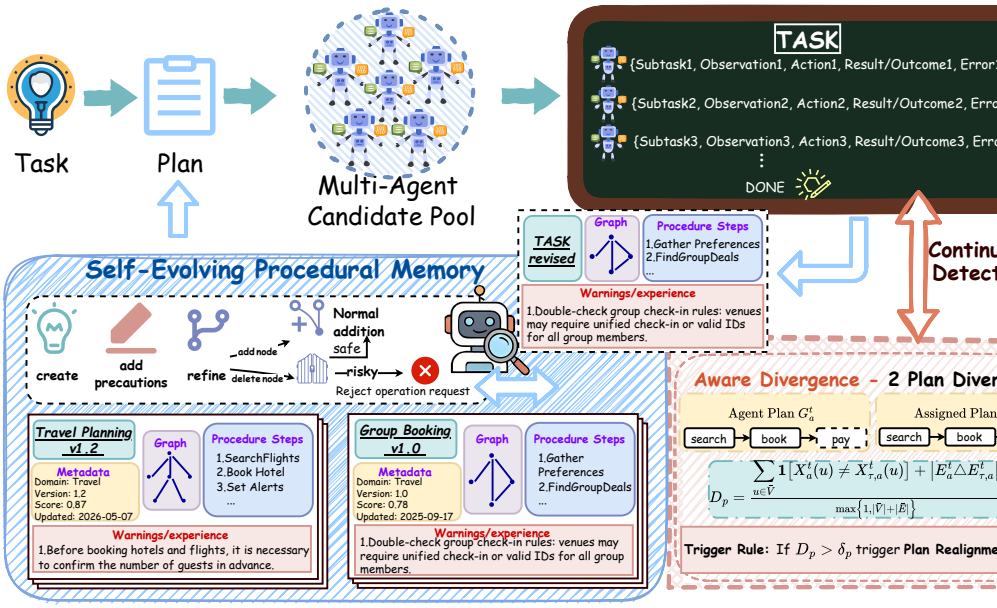
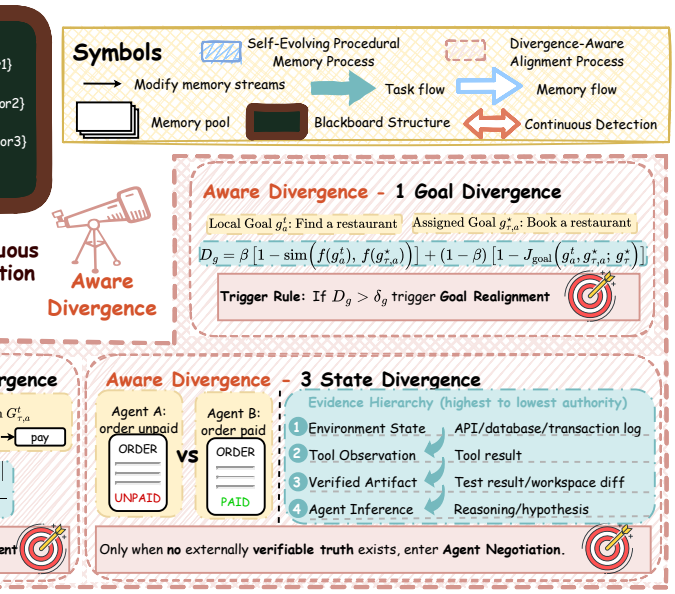

Task
Plan
Multi-Agent Candidate Pool
TASK
{Subtask1, Observation1, Action1, Result/Outcome1, Error
{Subtask2, Observation2, Action2, Result/Outcome2, Erro
{Subtask3, Observation3, Action3, Result/Outcome3, Erro
DONE
Continu
Detect
Self-Evolving Procedural Memory
TASK revised
Graph
Procedure Steps
1.Gather Preferences
2.FindGroupDeals
...
Warnings/experience
1.Double-check group check-in rules: venues may require unified check-in or valid IDs for all group members.
create
add precautions
refine
add node
delete node
Normal addition
safe
risky
Reject operation request
Travel Planning v1.2
Metadata
Domain: Travel
Version: 1.2
Score: 0.87
Updated: 2026-05-07
Graph
Procedure Steps
1.SearchFlights
2.Book Hotel
3.Set Alerts
...
Warnings/experience
1.Before booking hotels and flights, it is necessary to confirm the number of guests in advance.
Group Booking v1.0
Metadata
Domain: Travel
Version: 1.0
Score: 0.78
Updated: 2025-09-17
Graph
Procedure Steps
1.Gather Preferences
2.FindGroupDeals
...
Warnings/experience
1.Double-check group check-in rules: venues may require unified check-in or valid IDs for all group members.
Aware Divergence - 2 Plan Diver
Agent Plan G_a^t
search
book
pay
Assigned Plan
search
book
$D_p = \frac{\sum_{u \in \tilde{V}} \mathbf{1}[X_a^t(u) \neq X_{\tau,a}^t(u)] + |E_a^t \triangle E_{\tau,a}^t|}{\max\{1, |\tilde{V}| + |\tilde{E}|\}}$
Trigger Rule: If $D_p > \delta_p$ trigger Plan Realignme


Symbols
Self-Evolving Procedural Memory Process
Divergence-Aware Alignment Process
Modify memory streams
Task flow
Memory flow
Memory pool
Blackboard Structure
Continuous Detection
Aware Divergence
Aware Divergence - 1 Goal Divergence
Local Goal g_a^t: Find a restaurant
Assigned Goal $g_{\tau,a}^*$: Book a restaurant
$D_g = \beta\left[1 - \mathrm{sim}\left(f(g_a^t),\ f(g_{\tau,a}^*)\right)\right] + (1-\beta)\left[1 - J_{\mathrm{goal}}\left(g_a^t, g_{\tau,a}^*; g_\tau^*\right)\right]$
Trigger Rule: If $D_g > \delta_g$ trigger Goal Realignment
Aware Divergence - 3 State Divergence
Agent A: order unpaid
ORDER
UNPAID
vs
Agent B: order paid
ORDER
PAID
Evidence Hierarchy (highest to lowest authority)
1 Environment State — API/database/transaction log
2 Tool Observation — Tool result
3 Verified Artifact — Test result/workspace diff
4 Agent Inference — Reasoning/hypothesis
Only when no externally verifiable truth exists, enter Agent Negotiation.